

Exercise 10: Multi-Party Computation (MPC)

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Exercise 1: Boolean Secret Sharing

Let $a \in \{0, 1\}$ be a secret bit. In Boolean secret sharing, a is represented as

$$a = a_1 \oplus a_2.$$

- (a) Suppose $a = 1$. Give two different valid pairs (a_1, a_2) .
- (b) Explain why neither a_1 nor a_2 alone reveals any information about a .
- (c) If Alice holds a_1 and Bob holds a_2 , how can they reconstruct a ?

Exercise 2: XOR vs AND

Let's write $a = a_1 \oplus a_2$ and $b = b_1 \oplus b_2$. Assume that Alice holds shares a_1, b_1 and Bob holds shares a_2, b_2 .

- (a) Show how Alice and Bob can compute $a \oplus b$ without any communication.
- (b) Expand $(a_1 \oplus a_2) \wedge (b_1 \oplus b_2)$.
- (c) Explain why computing $a \wedge b$ requires communication.

Exercise 3: AND using Beaver Triple

Assume Alice and Bob want to compute

$$c = a \wedge b$$

using a Beaver triple (i, j, k) such that

$$k = i \wedge j.$$

All values are Boolean and XOR secret-shared.

- (a) Define the masked values

$$d = a \oplus i, \quad e = b \oplus j.$$

Explain why revealing d and e does not leak information about a or b .

- (b) Show that

$$a \wedge b = k \oplus (d \wedge j) \oplus (e \wedge i) \oplus (d \wedge e).$$

- (c) In the protocol, only Bob adds the term $(d \wedge e)$ to his share. Explain why this still gives a correct secret sharing of c .

Exercise 4: Matching Problem

Alice likes Bob if $a = 1$, and Bob likes Alice if $b = 1$. They want to know whether they like each other without revealing their individual preferences.

- (a) Which Boolean function represents mutual liking?
- (b) Describe how this function can be computed using Boolean secret sharing and a Beaver triple.
- (c) If the result is 0, what can Bob learn about Alice's preference?

Exercise 5 (Optional): Security Intuition

- (a) In the AND protocol, Alice and Bob reveal $d = a \oplus i$ and $e = b \oplus j$. What property of i and j ensures security?
- (b) Why is the Beaver triple generated in an offline preprocessing phase?

Solution to Exercise 1: Boolean Secret Sharing

- (a) If $a = 1$, valid secret sharings include for example:

$$(a_1, a_2) = (0, 1) \text{ or } (1, 0),$$

since $0 \oplus 1 = 1 \oplus 0 = 1$.

- (b) Each share a_1 or a_2 is uniformly random in $\{0, 1\}$. Therefore, observing a single share gives probability

$$\Pr[a = 0 \mid a_1] = \Pr[a = 1 \mid a_1] = \frac{1}{2},$$

so no information about a is leaked.

- (c) Alice and Bob reconstruct a by sending each value each other and computing

$$a = a_1 \oplus a_2.$$

Solution to Exercise 2: XOR vs AND

- (a) Each party locally computes its share of $a \oplus b$:

$$(a \oplus b)_1 = a_1 \oplus b_1, \quad (a \oplus b)_2 = a_2 \oplus b_2.$$

Then

$$(a \oplus b)_1 \oplus (a \oplus b)_2 = a \oplus b.$$

No communication is needed.

- (b) We expand:

$$(a_1 \oplus a_2) \wedge (b_1 \oplus b_2) = (a_1 \wedge b_1) \oplus (a_1 \wedge b_2) \oplus (a_2 \wedge b_1) \oplus (a_2 \wedge b_2).$$

- (c) The cross terms (e.g. $a_1 \wedge b_2$) involve secret values held by different parties. Hence communication is required to compute $a \wedge b$.

Solution to Exercise 3: AND using Beaver Triple

- (a) The values

$$d = a \oplus i, \quad e = b \oplus j$$

are masked by uniformly random bits i, j . Thus d and e are statistically independent of a and b , and revealing them leaks no information.

- (b) Using Boolean algebra:

$$\begin{aligned} a \wedge b &= (d \oplus i) \wedge (e \oplus j) \\ &= (d \wedge e) \oplus (d \wedge j) \oplus (e \wedge i) \oplus (i \wedge j) \\ &= k \oplus (d \wedge j) \oplus (e \wedge i) \oplus (d \wedge e), \end{aligned}$$

where $k = i \wedge j$.

- (c) Only Bob adds $(d \wedge e)$ to his share. Since boolean secret sharing only requires the XOR of both shares to be correct, this asymmetric assignment still results in a correct sharing of $c = a \wedge b$.

Solution to Exercise 4: Matching Problem

- (a) Mutual liking is represented by the Boolean function

$$a \wedge b.$$

- (b) Alice and Bob secret-share a and b . Using a Beaver triple (i, j, k) , they compute $a \wedge b$ via the AND protocol described in Exercise 3.
- (c) If $a \wedge b = 0$ and Bob knows $b = 1$, then Bob can conclude that $a = 0$. This information leakage is inherent and allowed, since MPC only hides information beyond the output.

Solution to Exercise 5 (Optional): Security Intuition

- (a) The security relies on i and j being uniformly random and unknown to the other party. This randomness perfectly masks a and b .
- (b) Beaver triples are generated offline because their production is expensive. Separating pre-processing from the online phase allows efficient secure computation during execution.